



DAYTONA HACKSPRINT SEOUL · SOLO BUILD

GateTrace

Research inputs verified before use,
with evidence preserved for every verdict

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Input errors propagate into research results



Contaminated inputs weaken analysis and training, while unexplained blocking discards valid evidence.

Researchers need to know what was checked and why an input was blocked before they use it.



Example users · Experimental data · Survey data · Training data



Pass or fail alone is not enough for research

01 Check scope The applied rules are unclear	02 Run environment Inputs and verification code are mixed	03 Failure reason Evidence is scattered across tools	04 Reproducibility The same run is difficult to trace
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| A verdict must travel with its execution evidence.



Researchers only choose a goal and input

01 State the goal Describe the intended research use	02 Select a sample Demo: clean or contaminated sensor CSV	03 Run in isolation Evaluate constrained gates in Daytona	04 Review evidence Verdict, gate evidence, and trace keys
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APPROVED or QUARANTINED, with the reason on one screen.



A contaminated input is quarantined with evidence

A • INPUT

Contaminated sensor CSV

Tabular-data implementation example

B • VERDICT

QUARANTINED

Three of four gates failed

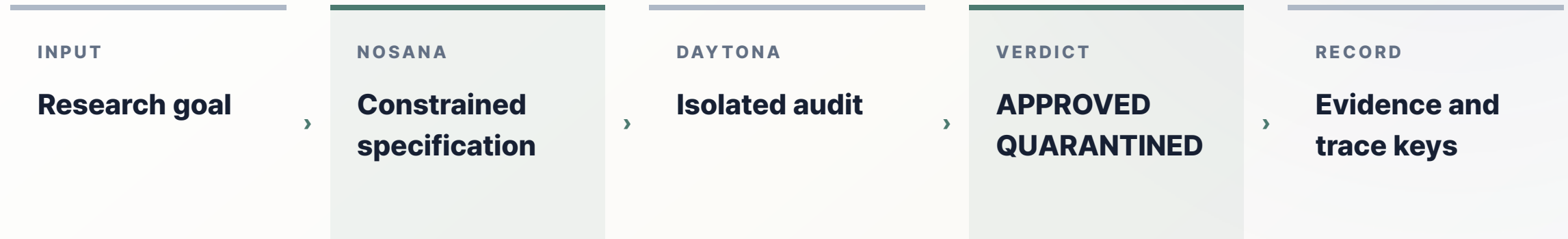
C • DAYTONA

26.217s

Sandbox ID and duration recorded



Generation is constrained; the verdict stays deterministic



Nosana

Maps the research goal to four allowed gates and explanations

Daytona

Runs the host-owned verifier in isolation, returns results, and deletes the sandbox

SAFETY BOUNDARY Nosana cannot change Daytona's deterministic verdict.



06 · VERIFIED RESULTS

Live results and reproduction paths are public

LOCAL CONTRACT

18 / 18 tests passed external services mocked

DAYTONA LIVE

Clean APPROVED 4/4 · contaminated QUARANTINED 1/4

VERCEL LIVE

gatetrace-daytona.vercel.app

PUBLIC REPOSITORY

github.com/geondongkim/gatetrace-daytona

Nosana live confirmed · qwen/qwen3.8-27b · Daytona clean/contaminated runs and sandbox deletion verified

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EXPECTED IMPACT

A verifiable adoption gate for research inputs

01

Block contamination

Quarantine failed inputs before analysis or training

02

Review every verdict

Preserve the request, run, and gate evidence together

03

Extend adoption rules

Apply the same gates to later datasets

NEXT STAGE

Evidence-preserving augmentation · retain provenance and accept only generated samples that pass the gates

This is an expected impact and extension plan. Document inputs require a dedicated verifier.